Adam **O'Camb**

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Experience

Wasco Switches and Sensors

Santa Maria, CA

Design Engineer and Project Manager

Jan. 2022 - Present

- Lead mechanical design, prototyping, validation, and manufacturing support for ultra-high-purity electromechanical pressure switches used in demanding gas delivery applications.
- Lead development of a high-temperature pressure switch, coordinating suppliers, manufacturing, quality, and operations to deliver production-ready samples on schedule while using less than 50% of the allocated budget.
- Designed, prototyped, tested, and manufactured a novel pressure switch in 5 months, delivering customer samples at least 50% faster than our standard development schedule.
- Designed, built, programmed, tested, and validated automated manufacturing and test equipment using LabVIEW, NI DAQ hardware, sensors, pneumatics, plumbing, electrical systems, custom machined mechanical hardware, and operator-facing HMIs. Doubled manufacturing throughput in some cases.
- Developed a spring database and automated data-entry workflow to standardize our test methods, organize our force-displacement data, characterize spring behavior, and improve spring selection for future pressure switch designs.
- Develop CAD models, assemblies, production drawings, and tooling designs in Autodesk Inventor; review designs for DFM/DFA, tolerance stack-up, precise fits, threaded interfaces, and appropriate GD&T.
- Plan and execute design validation testing, experiments, cyclic durability testing, root-cause investigations, and PFMEA/DFMEA risk assessments to improve product reliability and manufacturing readiness.
- Identified and resolved high-temperature switch reliability issues through design and process changes, reducing scrap by at least 20% while also decreasing cycle time.
- Designed, built, programmed, tested, and validated an automated cleaning station integrating pneumatic, hydraulic, electrical, and control systems. Improved cycle time by over 200% while maintaining ultra-high-purity cleanliness standards and improving operator usability.
- Use 3D printing, CAM, CNC milling and turning, manual machining, and shop equipment to rapidly prototype designs, fabricate fixtures and tools, repair equipment, and support manufacturing needs.
- Create procedures, manufacturing work instructions, test documentation, and project updates; present technical findings and progress to engineering leadership, executives, and cross-functional teams.

iFixit

San Luis Obispo, CA

Senior Technical Writer

May 2017 - Jan. 2022

- Reverse-engineered consumer electronics to understand product architecture, failure modes, assembly methods, and repair procedures, then translated findings into clear technical documentation for a global user base.
- Led repairability assessment work for corporate clients, providing design-for-repairability feedback to major consumer electronics manufacturers.
- Trained and mentored new team members in repair technique, technical writing, photography, documentation standards, and company culture.

American Robotics Company

San Luis Obispo, CA

Mechatronics Engineer

Jan. 2017 - Feb. 2017

- Designed, modeled, tested, troubleshot, repaired, and maintained rugged robotic systems in a fast-paced startup environment.
- Created SolidWorks models and drawings for manufacturing partners.

Education

California Polytechnic State University

San Luis Obispo, CA

B.S. in Mechanical Engineering, Concentration in Mechatronics

2012 - 2016

- Mechatronics, System Dynamics, Control Systems, Assembly, C++, MicroPython, Fluid Dynamics, Heat Transfer

Applicable Skills

Technology

Solidworks CAD and FEA (CSWA), AutoDesk Inventor CAD and CAM, Fusion 360, Git, LabVIEW, basic Python, KiCAD, Cursor

Equipment

Shop tools, manual and CNC mill and lathe, soldering, oscilloscope, data acquisition system, 3D printer, NI DAQ hardware, sensors

Activities and Involvement

Level 2 NICA Certified Coach, NICA Team - San Luis Obispo Scoters — *2021 - 2025*

Category 1 XC Racing and Category 2 Road Racing, Various Teams — *2009 - Present*

Personal Cycling Coach, Freelance — *2023 - Present*